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Understanding the role of study strategies and learning disabilities on student academic performance to enhance educational approaches: A proposal using artificial intelligence

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ABSTRACT

Statement of problem: The students' academic performance is influenced by a complex interplay among several factors. Traditional educational approaches often struggle to accommodate the diverse needs of students, leading to suboptimal learning outcomes.

Purpose: This article aims to comprehensively understand the role of study strategies and learning disabilities in shaping academic performance. Through the integration of artificial intelligence (AI) tools, the purpose is to propose a decision support system (DSS) for recommendations to improve the educational approach.

Method: To identify features with higher explanatory power based on empirical data, we employed an artificial neural network (ANN) to recognize patterns of association between study strategies, learning disabilities, and academic performance. Using the pondered features, a Fuzzy-based AI was built for offering recommendations into effective educational interventions.

Conclusions: The findings underscore the significance of study strategies in mitigating the negative impact of learning disabilities on academic performance. By leveraging the proposed AI tools framework, educators can make informed decisions to tailor educational approaches, catering to the unique cognitive profiles of students. Personalized interventions based on identified patterns can lead to improved academic outcomes and greater inclusivity in the learning environment.

Practical implications: Educators and policymakers can adopt the proposed data-driven strategies to enhance teaching methodologies, thereby accommodating the varying needs of students with learning disabilities. This approach fosters a more inclusive and equitable educational landscape, promoting academic success for all learners.

1. Introduction

How can study strategies integrated into artificial intelligence (AI) tools mitigate the impact of learning disabilities on academic performance? Achieving optimal academic performance is a multifaceted challenge that requires a profound understanding of study strategies and their interaction with learning disabilities. The pursuit of effective educational approaches that cater to the diverse needs of students remains an ongoing endeavor, marked by the ever-evolving integration of computational advancements.

The educational trajectory is characterized by individual learning paths influenced by a myriad of factors, among which study strategies play a pivotal role. Simultaneously, a subset of students grapples with learning disabilities, which can significantly impede their academic progress. Traditional educational models, while indispensable, often struggle to accommodate the unique cognitive profiles and requirements of these students. This calls for innovative approaches that harness emerging AI-based technologies to bridge the gap between cognitive

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diversity and educational outcomes.

Extensive research has highlighted the importance of study strategies in shaping learning experiences and academic achievement. However, the nuanced relationship between these strategies and the presence of learning disabilities using AI tools demands further exploration. Existing studies have laid the groundwork by identifying key factors that influence academic performance, but a comprehensive understanding of how study strategies can mitigate the impact of learning disabilities remains an underexplored territory.

The current body of literature presents a conspicuous research gap pertaining to the integration of AI in analyzing the interaction between study strategies, learning disabilities, and academic performance. While studies have highlighted the potential of AI in personalized learning, its precise role in optimizing educational approaches for students with learning disabilities requires more comprehensive investigation.

This article aims to investigate the intricate relationship between study strategies, learning disabilities, and academic performance. Utilizing AI as a tool for informed decision-making, our objective is to propose effective enhancements in educational approaches that facilitate personalized learning experiences. This, in turn, optimizes academic achievement for students with diverse cognitive profiles and addresses the challenges posed by learning disabilities.

Thus, this study seeks to address the aforementioned research gap by undertaking an examination of the interplay between key factors. By harnessing the power of AI algorithms, we aim to identify patterns and correlations that can inform evidence-based decision-making in educational contexts. The proposed research aspires to offer practical insights for educators, administrators, and policymakers, paving the way for the design and implementation of targeted interventions that enhance academic outcomes for all students, regardless of their cognitive profiles or learning challenges. Through this multidimensional exploration, our study endeavors to contribute to the advancement of educational practices that prioritize inclusivity, equity, and excellence.

To address the initial research question, this paper is organized into key sections. The first, "Related Works," delves into existing research, analyzing prior studies to establish the contextual underpinning for our investigation. Following this, the section titled "AI-based Decision Support System Modeling" elucidates the methodology behind creating the model, outlining its framework and the integration of constituent elements. Subsequently, the "Application of the Proposed Framework" section provides practical validation through a case study, accompanied by an evaluation of the resultant proposed system. This effort allows us to showcase the concrete applicability and efficacy of the DSS in addressing challenges associated with enhancing educational approaches. Lastly, the "Concluding Remarks" section encapsulates the study's discoveries and their broader implications.

2. Related works

To search for the most reliable, relevant, and up-to-date related works, Science Direct database were used, a premier source for scientific research. The following query was applied: title-abs-key ("education" or "study strategy" or "learning disability" or "student performance") and ("artificial intelligence" or "machine learning" or "computational tool") and (limit-to (pubstage, "final")) and (limit-to (doctype, "ar")) and limit-to (subjarea, "edu")) and (limit-to (pubyear, "from 2014 to 2023")) and (limit-to (language, "english")). This search yielded 204 published works, which were preliminarily screened. After excluding those outside the scope, 35 articles were selected and discussed below.

2.1. Inclusive education for students with learning disabilities

Students facing learning disabilities confront obstacles across various academic domains, encompassing phonological awareness, language comprehension, time management, information processing, and problem-solving skills. What study strategies are most effective for students with learning disabilities? Previous studies suggest that students with disabilities employ diverse learning strategies, while educators can implement accommodations and practices to support their learning effectively.

Alnahdi (2015) conducted an extensive literature analysis, focusing on instructional strategies targeting improved skills among students with learning disabilities. Studies were categorized into early reading methods, comprehensive approaches, and singular techniques, emphasizing the need for intensive, explicit, systematic instruction for reading skill enhancement. Maciver et al. (2018) present findings from educators identifying supportive strategies for learners with disabilities, emphasizing the role of environment and pedagogy. Bakken (2017) discusses mnemonic strategies to aid students with intellectual and developmental disabilities in memory retention.

Moriña Díez (2019) emphasizes the importance of emotions in the realm of education, particularly for students with disabilities, while also shedding light on the crucial aspect of faculty-student relationships. The study delves into the various strategies adopted by students with disabilities to facilitate their learning process, reflecting a wide range of approaches. The author further highlights the motivational tactics employed by educators to cultivate a sense of inclusion, thereby enriching the overall learning journey. Andreou, Athanasiadou, and Tzivinikou (2019) discovered that accommodations such as extended time and read-aloud assistance effectively enhanced reading comprehension assessments for such students.

Mogan and Yasin (2020) engaged in a qualitative study to gather professors' insights on selecting and implementing teaching strategies, identifying that multisensory strategies, interactive multimedia, and group mentoring were effective for teaching reading skills to students with learning disabilities. Vats and Dey (2022) delve into classroom accommodation strategies, enhancing disabled students' academic journey by cultivating empathy among educators and providing meaningful assistance. The study also addresses variables influencing accommodations and their impact on disabled students' mental well-being.

In the face of this challenge, how can AI tools be used to help students with learning disabilities? Overall, AI techniques can contribute by replicating human decision-making, minimizing uncertainties, and enhancing learning contexts, emphasizing their importance for intelligent and adaptable e-learning environments.

Yahya and Noor (2015) propose an e-learning DSS that estimates students' learning styles automatically, making learning easier and enhancing progress. Colchester, Hagras, Alghazzawi, and Aldabbagh (2017) discuss the use of AI techniques to develop accurate student profiles and models for adaptive learning environments. Perelmutter, McGregor, and Gordon (2017) provide a systematic review of assistive technology interventions for students with learning disabilities, finding that while assistive interventions proved overall beneficial, the most effective interventions were based on word processing, multimedia, and hypertext. Speech-to-text systems had a small positive effect, while smart pens and text-to-speech systems presented mixed results.

Interactive instruction models were found to positively impact students, promoting educational and social inclusion, as shown by García-Carrión (2018). In this sense, Naumova, Vytovtova, Mitiukov, and Zulfugarzade (2017) developed e-learning methods for students with disabilities, and Hedin and Kann (2019) demonstrated that a course integrating study skills and reflection seminars enhanced students' ability to analyze and adjust their study habits. Cukurova, Kent, and Luckin (2019) argue that AI can support human decision-making processes in education rather than replacing them through automation. Banerjee, Lamrani, Hossain, Paudyal, and Gupta (2020) developed an AI tutor that can teach users gestures required on the field as a prerequisite, reducing resource requirements for training and decreasing the time taken for making personnel job-ready.

Martiniello et al. (2021) point out that the current decade reflects on the potential of AI-based apps to facilitate learning for all post-secondary

students, including those with disabilities. According to August and Tsaima (2021), AI and ML algorithms can enhance learning systems, acting as metaphorical support systems for instructors. These tools may empower educators to offer better guidance while students enjoy increased autonomy, supporting automated tutoring, personalized learning, knowledge assessment, and task automation.

Tiwari (2023) highlights the potential of AI-assisted systems for personalized learning, adaptive testing, intelligent tutoring systems, learning analytics, and content creation. From a literature review, the author explores AI integration in education, aiming to enhance personalized learning experiences. Emerging areas like virtual reality education and educational game design also show promise. Overall, AI's potential to enhance learning is significant, but ethical considerations and continued research are crucial.

2.2. AI-based models for enhancing educational approaches

Kolo and Adepoju (2015) focus on using a decision tree (DT) approach to predict students' academic performance, with factors such as financial status, motivation to learn, and gender found to affect performance. Thus, the study aimed to help students and teachers improve student performance and reduce the failing ratio by taking appropriate steps at the right time. However, the research only considers a limited set of factors such as financial status, motivation to learn, and gender, and does not explore other potential factors that may influence students' academic performance.

Mesarić and Šebalj (2016) create a model that classifies students based on their academic success at the end of their first year, using data on high school success, courses taken, and rank of faculty preferences. The model uses DT algorithms and identifies significant variables such as total points in the state exam, points from high school, and points in the Croatian language exam. While the Reduced Error Pruning Tree (RepTree) algorithm achieved the highest accuracy of 79%, the J48 algorithm demonstrated a higher overall classification rate. The study highlights the importance of predicting student success in higher education and provides insights for future research. However, the authors acknowledge that the model's accuracy rate could be improved by introducing additional variables, such as course grades on the first year of study or conducting primary research among the student population.

Miguéis et al., 2018 propose a two-stage model that uses data mining techniques to predict the overall academic performance of university students based on their first-year academic career. It considers both the average grade achieved and the time taken to complete the degree as indicators of academic success. The study found that Random Forest (RF) model was a superior classification technique compared to DT, Support Vector Machine (SVM), Naive Bayes, bagged trees, and boosted trees. The model and segmentation framework can help institutions promote higher performance levels and mitigate academic failure, ultimately enhancing the quality of the academic experience provided by higher education institutions.

Buenaño-Fernández et al., 2019 introduce an application of ML technique based on DT to predict the final grades of computer engineering students based on their historical performance of grades. The methodology involves data collection, pre-processing, grouping of students with similar performance patterns, selection of a supervised learning algorithm, and experimental process. The results demonstrate the effectiveness of the DT technique in predicting student performance. The model can be applied to support action plans to prevent dropouts, personalize student follow-up, and provide valuable information to students for evaluating their academic performance. This includes taking improvement actions in subjects with a higher risk of failing.

Waheed et al. (2020) focus on using deep learning models to predict the academic performance of students based on data from virtual learning environments (VLEs). The study uses a deep artificial neural network on handcrafted features extracted from clickstream data to predict at-risk students and provide measures for early intervention. The proposed model achieves a high accuracy, outperforming logistic regression (LR) and SVM models. The study aims to assist institutes in formulating a framework for pedagogical support and facilitating decision-making in higher education, but in addition to ANN-based models being black boxes with poor interpretability, the study does not provide a clear demarcation line or improved accuracy between quartiles for the 'distinction-pass' category, which suggests a limitation in accurately predicting the performance of distinction students.

Cruz-Jesus et al. (2020) presents a novel approach using ML techniques to predict academic achievement in public high schools in Portugal, comparing the performance of different AI methods. The accuracy results show that k-nearest neighbors (KNN), RF, and DT had the best performers. Artificial Neural Networks (ANN) and Extremely Randomized Trees (ERT) also achieved high accuracy, but SVM performed poorly. The paper does not explore the interpretability of these models, which could be important for making informed decisions in the education sector. The authors acknowledge that the AI models used in the study are essentially black boxes, providing no information about how they arrived at their predictions. This limitation could restrict their widespread acceptance and adoption. Moreover, the study focuses on academic achievement in public high schools in a specific location, and don't provide mechanisms to adaptation of model to other contexts or conditions, which may limit the generalizability of the findings to other educational contexts or countries.

Lázaro Alvarez, Callejas, and Griol (2020) present a case study on predicting potential dropout in Computer Engineering in Cuba using institutional data and classification features obtained at the time of enrollment and after the first semester and academic year. The authors tested a combination of classification features for training and testing DT (J48) and Multilayer Perceptron (MLP), being the best performance obtained with the latter. The results show that it is possible to obtain an early assessment of the risk of dropout, with high accuracy and recall for non-promoting students, which can help in defining prevention policies. The study primarily relies on decision trees and neural networks for prediction but does not explore the potential limitations or drawbacks of these specific methods in predicting student dropout.

Rodríguez-Hernández et al., 2021 tested a systematic procedure for implementing an ANN to predict academic performance in higher education and analyze the importance of various predictors of academic performance. ANNs outperform other machine-learning algorithms. Additionally, prior academic achievement, socioeconomic conditions, and high school characteristics are identified as important predictors of academic performance in higher education. The study acknowledges that the predictor of prior academic achievement was operationalized only using test outcomes, and any other information on students' grades from high school was available. This means that relevant information represented by high school grade point average could not be included in the analysis.

Riestra-González et al., 2021 analysis the early prediction of academic performance, with the objective is to detect at-risk, fail, and excellent students in the early stages of the course. Different classification models, including DT, Naive Bayes, LR, MLP neural network, and SVM, are created and evaluated. The accuracies of these models improve as the moment of prediction increases, with MLP achieving the best performance. The study focuses on solving learning management system issues and does not consider other aspects of student performance.

Mansouri, Ravasan, and Ashrafi (2021) discuss the use of the Learning Fuzzy Cognitive Map (LFCM) approach for predicting student performance. As an advantage, the authors highlight the limitations of common prediction approaches like ANN and LR in dealing with small sample sizes, and propose the LFCM model as a practical alternative, which performs well on limited sample sizes and can be easily interpreted by humans. The research was conducted on a master's program, collecting longitudinal data from 30 postgraduates in four subsequent semesters, but does not explicitly provide detailed empirical findings regarding an overall accuracy of the model, focusing on each student

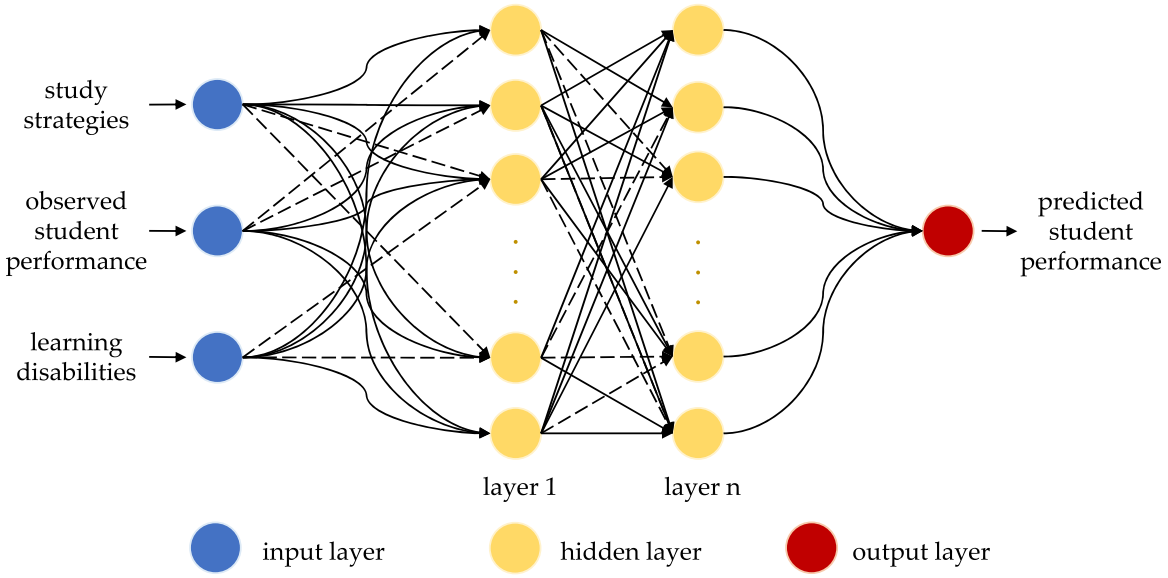


Fig. 1. ANN based on MLP deep learning architecture.

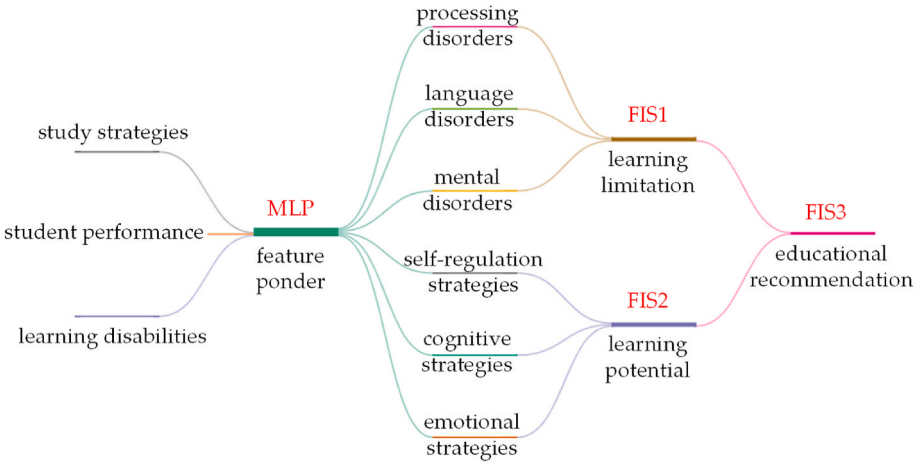


Fig. 2. AI-based decision support system framework.

separately to construct his/her own LFCM regarding performance antecedents and performance for three semesters, using the proposed algorithm. To such these analysis condition, the model achieved a good performance of the predicting, having in mind the small sample size. Jayagowri and Karpagavalli (2021) also propose the use of LFCM technology to predict students' performance. The model showed accurate predictions with small sample sizes. The authors point that LFCM model proposed in their study may require further fine-tuning of its components to improve its accuracy and effectiveness, and there may be other factors that were not considered in the model that could also impact student performance.

Petra and Aziz (2021) focus on analyzing student performance in higher education using fuzzy logic evaluation methods. A fuzzy inference system (FIS) based on the Mamdani technique was designed and tested on real students' mark samples. The results of the fuzzy expert system and conventional methods were compared to identify deficiencies in academic or personality development. As an advantage highlighted by the authors, the fuzzy logic provides flexibility in allocating different weightages to each attribute based on the needs of the education institution. Thus, the proposed FIS can help in investigating and analyzing student performance, enabling educators to identify deficiencies and advantages of students, and assist higher education institutions in determining appropriate lectures, activities, and assessments for students. The paper lacks information on the sample size or diversity of the students used in testing the FIS. In addition, any discussion regarding the generalizability of the findings and whether the results can be applied to different educational contexts or institutions are not presented.

Niyogisubizo, Liao, Nziyumva, Murwanashyaka, and Nshimyumukiza (2022) propose a novel stacking ensemble machine learning approach to predict student dropout in university classes based on a hybrid of RF, Extreme Gradient Boosting (XGBoost), Gradient Boosting (GB), and Feed-forward Neural Networks (FNN). The proposed method demonstrates greater performance compared to the base models, as evaluated by testing accuracy and the area under the curve (AUC) metrics. The findings suggest that the proposed method can identify students at risk of dropping out and enable early intervention to prevent dropout. As a limitation, while the proposed approach demonstrates better performance compared to the base models, the paper does not provide a detailed analysis of the interpretability of the model. The black-box nature of some machine learning algorithms used in the ensemble may limit the ability to understand the underlying reasoning

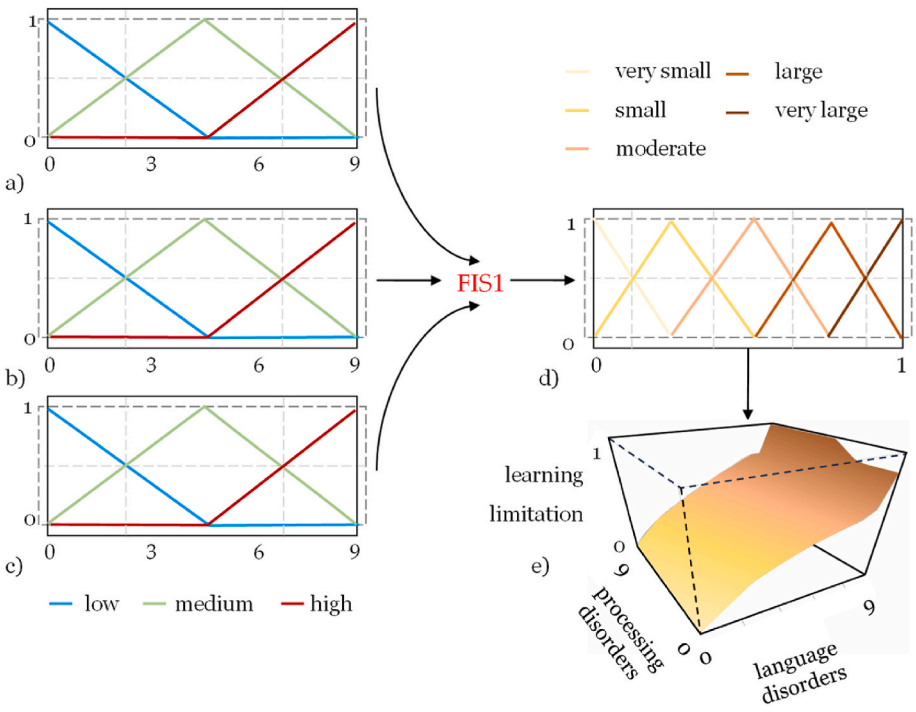


Fig. 3. FIS1: (a) processing disorders; (b) language disorders; (c) mental disorders (d) learning limitation; (e) decision surface.

Table 1
Learning limitation based on rule basis of FIS1.

processing	language	mental		
		low	*medium*	*high*
low	*small*	very small	very small	small
	medium	small	small	small
	large	small	small	moderate
medium	*small*	small	moderate	moderate
	medium	moderate	moderate	moderate
	large	moderate	moderate	large
high	*small*	moderate	moderate	large
	medium	moderate	moderate	large
	large	large	large	very large

process and interpret the model for humans, which may harm greater acceptance and application of the proposed model by professionals in the education sector.

Bressane et al. (2022) propose a fuzzy artificial intelligence-based model to forecast student performance and retention risk in engineering education. The model is built using historical records of students' grades and retention, as well as data collected through an online survey on learning strategies. By using data based on self-reported data collected through an online survey, which may be subject to inaccuracies, the main challenge of the study was to propose a framework capable of providing accurate predictions, dealing with uncertainties inherent in small data and subjective data based on the student perception. The proposed model outperformed other AI methods, including Cascade-Correlation Network (CCN), MLP, Polynomial Neural Network (GMDH), Probabilistic Neural Network (PNN), Radial Basis Function Network (RBFN), Gene Expression Programming (GEP), Decision Tree Forest (Tree Boost), and SVM. The model's practical implication is to support students' planning and assist professors in identifying students at higher risk of retention, enabling preventive interventions. As the fuzzy model is based on a set of easily understood rules, its interpretability is a huge advantage over other AI models, which are a real black box, so the proposed model should achieve greater acceptance by education professionals. Nevertheless, the proposal does not address factors beyond academic performance that may contribute to students' performance, such as health conditions or learning disabilities.

Priyambada et al. (2023) propose a two-layer ensemble learning technique that combines ensemble learning and ensemble-based progressive prediction to predict students' performance using their learning behavior and domain knowledge. The proposal showed improved accuracy on a real-world student dataset, outperforming SVM, KNN, and RF models. The authors argue that proposed framework can assist academic stakeholders in taking actions to ensure timely graduation of students and provide valuable information for students to make informed decisions about their academic paths. However, the framework does not encompass elements beyond academic achievement that could potentially influence students' outcomes, such as health conditions or impediments to learning.

Han (2023) focuses on the analysis of college students' stress levels, psychological well-being, and academic performance using the Heuristic Fuzzy C-means Clustering Algorithm (HFCA). The study aims to identify the most crucial aspects of student success, such as student involvement and satisfaction. The proposed HFCA method achieves high performance in the prediction ratios compared to other methods, but the model has limitations in performing with a limited-size dataset. The paper also emphasizes the importance of understanding risk factors for poor mental well-being and developing targeted interventions to prevent health problems among college students.

2.3. Research gap

The existing literature, as demonstrated through the comprehensive review of related works, has shed light on various facets of inclusive education, student performance prediction, and the potential of AI-driven tools. However, despite the advances, there remains a research gap that calls for a more holistic and integrated exploration of the intricate relationship between study strategies, learning disabilities, and their collective impact on academic performance. This research gap underscores the need for a study that bridges the existing knowledge and delves deeper into the interplay of these factors to propose practical

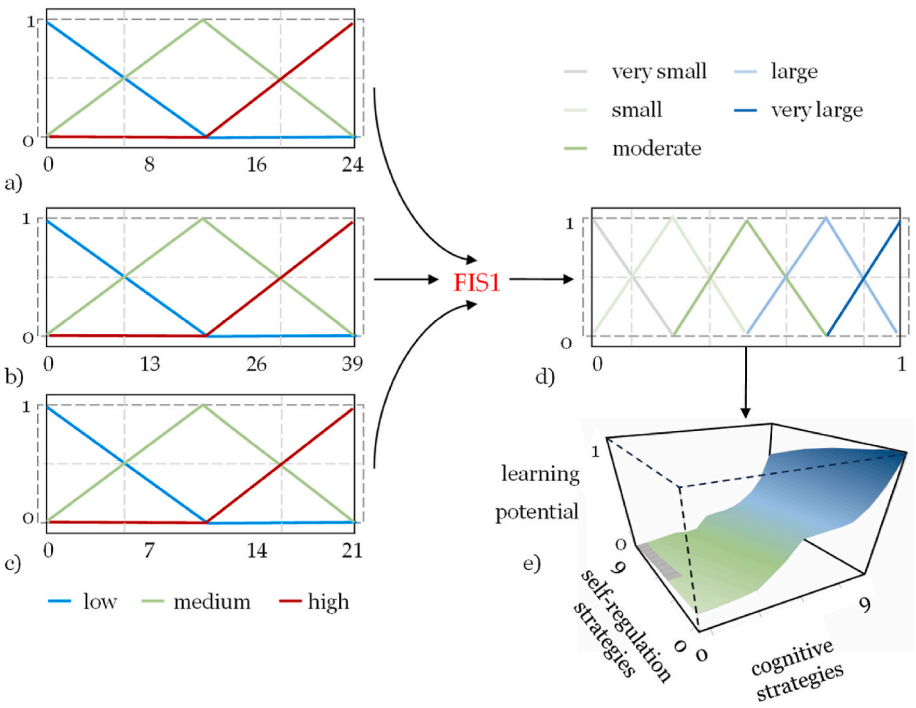


Fig. 4. FIS2: (a) self-regulation strategies; (b) cognitive strategies; (c) emotional strategies; (d) learning potential; (e) decision surface.

Table 2
Learning potential based on rule basis of FIS2.

emotional	cognitive	self-regulation		
		low	*medium*	*high*
low	*small*	very small	very small	small
	medium	small	small	small
	large	small	small	moderate
medium	*small*	small	moderate	moderate
	medium	moderate	moderate	moderate
	large	moderate	moderate	large
high	*small*	moderate	moderate	large
	medium	moderate	moderate	large
	large	large	large	very large

Table 3
Educational recommendation based on rule basis of FIS3.

		learning limitation		
		low	*medium*	*high*
learning potential	*low*	1	2	3
	medium	4	5	6
	high	7	8	9

solutions for enhancing the educational approach.

While previous studies have individually examined study strategies, inclusive education practices, and AI's potential in education, there is a dearth of research that synthesizes these elements into a unified framework. Moreover, the influence of learning disabilities on the effectiveness of study strategies and the subsequent academic performance is a relatively underexplored area. Such an investigation holds substantial significance given the increasing diversity of learners in educational settings and the evolving role of AI in educational technologies.

Thus, the present study aims to address this research gap, with a special focus on the potential of AI-driven tools to enhance these interactions. By integrating insights from multiple disciplines including education, psychology, and AI, this study seeks to provide a more nuanced and complete perspective on how students with learning disabilities engage with study strategies and how these interactions influence their academic outcomes.

Therefore, the motivation for conducting this study is multifaceted. First, it aligns with the current trends in educational research and practice that emphasize the importance of personalized learning experiences for all students, including those with learning disabilities. Second, the integration of AI tools into education is a burgeoning field, and there is a growing need to explore how these tools can be harnessed to provide meaningful support for students with diverse learning needs. Third, the study can contribute to the development of practical interventions and decision support systems that educators and institutions can utilize to enhance their inclusive educational practices.

In summary, the identified research gap, which stems from the limited integration of study strategies, learning disabilities, and AI in educational contexts, justifies the need for a comprehensive study that investigates these factors collectively. This study can provide valuable insights for educators, researchers, and policymakers aiming to create inclusive and effective learning environments for all students. Furthermore, the proposed decision support system has the potential to offer actionable recommendations, bridging theory and practice for the betterment of education.

3. AI – based decision support system modeling

The methodology employed in crafting a DSS to enhance education approach included the use of AI algorithms based on ANN and fuzzy inference system (FIS). To do this, an empirical dataset needs to cover key factors that influence students' academic performance, as demonstrated later, through a case study.

The collected data underwent thorough preprocessing to ensure uniformity and integrity. The data should be then structured into appropriate features, such as the frequency of specific study strategies, and indicators of how much the existence of learning disabilities hinders performance, and the final course grades [0, 10] achieved by the respondents. The study strategies are organized into three categories: 'self-regulation strategies,' 'cognitive strategies,' and 'emotional strategies.'

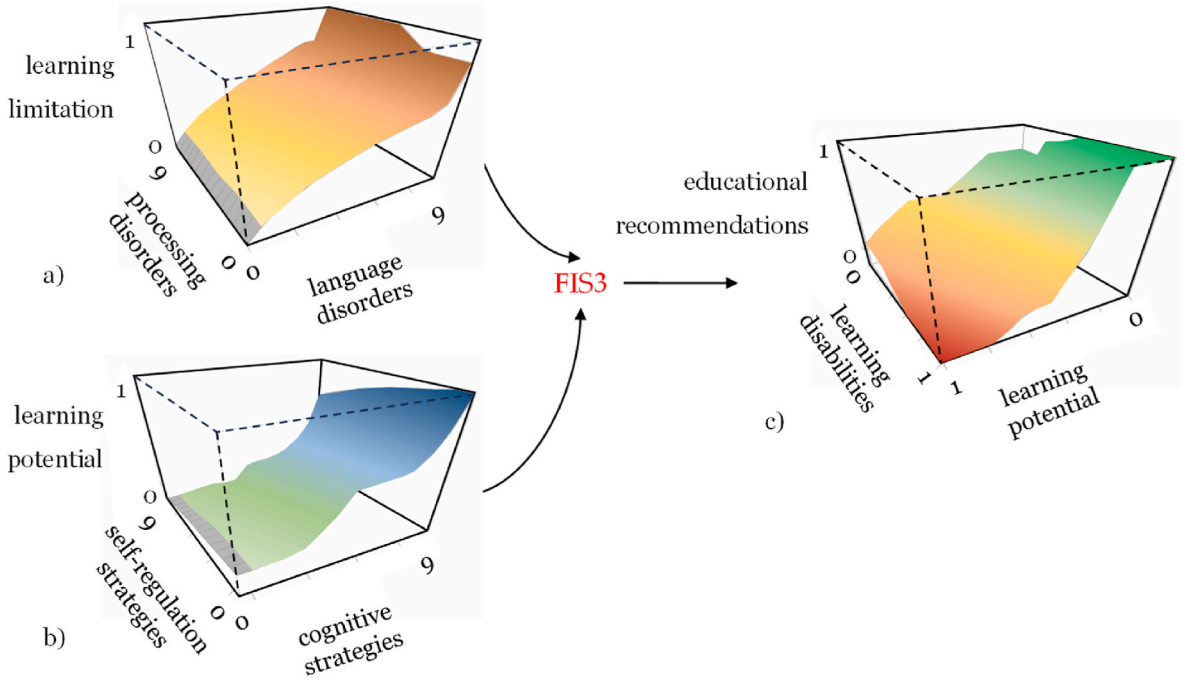


Fig. 5. FIS3: (a) learning limitation; (b) learning potential; (c) educational recommendation.

Within these categories, specific study strategies have been classified based on their characteristics and intended focus.

In this sense, the self-regulation strategies encompass 'Studying before class', 'Reviewing material prior to assessments', 'Consistent material review', 'Regular attendance in class', 'Seeking assistance when needed', 'Collaborative group study', 'Sustained attention during study sessions', 'Increased effort when encountering challenges'. In turn, the cognitive strategies includes 'Taking classroom notes', 'Summarizing lectures', 'Engaging in active reading', 'Creating reading summaries', 'Solving exercises for practice', 'Generating questions to enhance understanding', 'Constructing diagrams for visualization', 'Relating new knowledge to prior information', 'Mentally rehearsing concepts for reinforcement', 'Teaching others to solidify comprehension', 'Employing mnemonic techniques for memory enhancement', 'Utilizing audio recordings for review', 'Employing flashcards for quick recall'. Finally, the emotional strategies comprise 'Maintaining calmness when facing difficulties', 'Sustaining composure despite the possibility of subpar performance', 'Staying composed when making errors during exercises', 'Engaging in relaxation techniques for emotional equilibrium', 'Seeking motivation to combat waning interest', 'Reflecting on the broader relevance of content to other subjects', 'Contemplating the significance of content for one's future profession'.

Similarly, the learning disabilities are classified into three distinct groups: 'processing disorders', 'language disorders', and 'mental disorders'. This categorization aims to capture the varied dimensions of challenges that learners might encounter in their educational journey. The processing disorders encompass 'Phonological dyslexia', 'Dyscalculia', and 'Ageometric' difficulties. The language disorders include 'Visual dyslexia', 'Dysgraphia', and 'Dysnomia. The mental disorders comprise 'Anxiety', 'Stress', and 'Depression'.

Then, a deep learning MLP-based neural network is modeled with this preprocessed empirical dataset. Through 5-fold cross-validation, the MLP learns to recognize patterns related to associations between key factors (Fig. 1).

The ANN modeling can be performed using the Orange Data Mining software (Demsar et al., 2013), a platform that offers a range of tools for analyzing and modeling complex datasets. Specifically, the 'Explain' functionality within the software is crucial in the process of feature weighting by the ANN. This weighting is guided by the variables' contribution to the coefficient of determination (R^2), an indicator of the model's ability to explain the variance observed in the dependent variable. By assessing the influence of each variable on the R^2 value, the AI can discern the most influential factors driving academic performance.

Building upon the pondered features by the ANN, a model grounded in fuzzy logic is developed to serve as the DSS for recommendations for effective educational interventions geared toward enhancing students' academic performance. The choice to employ a fuzzy logic-based model derives from its innate capability to navigate and accommodate the inherent uncertainty intrinsic to subjective data, in this case, data based on respondents' perceptions (Bressane, Biagolini, Mochizuki, Roveda, & Lourenço, 2017; Bressane, Mochizuki, Caram, & Roveda, 2016). Fuzzy logic is particularly apt for processing data that embodies varying degrees of certainty or vagueness, which are commonplace when dealing with human-centric inputs (Bressane, Silva, P.M., Fiore, Carra, Ewbank, & De-Carli, 2020).

In the context of our study, the data obtained from the self-reporting forms carries a level of subjectivity inherent to the respondents' interpretations and self-assessments. Thus, fuzzy logic enabled the incorporation of subjectivity into the decision-making process, allowing us to effectively model the nuanced and diverse factors influencing academic performance. By adopting fuzzy logic, the DSS is equipped to represent the complexity of the cognitive strategies' effectiveness and the presence of learning disabilities as gradual and imprecise values. This flexibility enables the DSS to handle cases where the boundary between the effectiveness of strategies or the certainty of the presence of a learning disorder may not be clearly delineated. Fuzzy sets and fuzzy rules enable the incorporation of linguistic variables and the incorporation of qualitative insights into the decision-making framework (Bressane et al., 2023a,b; Loureiro, Bressane, Nascimento, Simões, & Negri, 2023). As such, the fuzzy logic-based model integrates both quantitative and qualitative data, allowing us to use the subjective perspectives of the respondents. This holistic approach empowers educators and administrators to make well-rounded decisions that account for the diversity of student experiences and perceptions. This proposed framework can be seen in Fig. 2.

The FIS is built by integrating two preceding subsystems. The first

Table 4
Educational approach recommendations based on FIS3.

crisp output	potential, limitation	fuzzy AI - based recommendations
1	small, small	(*i*) educational approach: focus on providing a supportive learning environment with personalized attention; (*ii*) teaching techniques: a blend of one-on-one mentoring and interactive discussions; (*iii*) assessment methods: frequent verbal assessments to gauge understanding and provide immediate feedback
2	small, medium	(*i*) educational approach: employ differentiated instruction to cater to varying learning speeds; (*ii*) teaching techniques: utilize multimedia resources and practical examples to enhance engagement; (*iii*) assessment methods: combine written quizzes with oral assessments to accommodate diverse learning styles
3	small, high	(*i*) educational approach: implement targeted interventions for specific learning disorders, integrating specialized support; (*ii*) teaching techniques: multisensory approaches, visual aids, and repetition for reinforcement; (*iii*) assessment methods: offer alternative assessment formats such as projects and presentations
4	medium, small	(*i*) educational approach: foster a student-centered approach with opportunities for exploration; (*ii*) teaching techniques: combine expository teaching with problem-solving exercises to enhance critical thinking; (*iii*) assessment methods: employ quizzes, class discussions, and collaborative projects
5	medium, medium	(*i*) educational approach: promote active learning by incorporating group activities and peer discussions; (*ii*) teaching techniques: utilize case-based learning to bridge theoretical concepts with real-world applications; (*iii*) assessment methods: use a mix of quizzes, presentations, and written assignments
6	medium, high	(*i*) educational approach: tailor interventions for specific learning limitations, offering additional support; (*ii*) teaching techniques: blend flipped classroom methods with guided discussions; (*iii*) assessment methods: incorporate diverse assessment formats such as role-playing, simulations, and reflective essays
7	high, small	(*i*) educational approach: encourage independent exploration and inquiry-based learning; (*ii*) teaching techniques: incorporate project-based learning and open-ended discussions to stimulate curiosity; (*iii*) assessment methods: implement self-assessment, peer review, and research-based projects to foster critical thinking
8	high, medium	(*i*) educational approach: foster a collaborative learning environment with opportunities for peer mentoring; (*ii*) teaching techniques: use technology-enhanced learning platforms and group problem-solving activities; (*iii*) assessment methods: combine quizzes, debates, and collaborative presentations
9	high, high	(*i*) educational approach: implement individualized education plans, offering comprehensive support; (*ii*) teaching techniques: project-based learning and flipped classroom methods; (*iii*) assessment methods: use a range of assessments including research papers, multimedia presentations, and group projects

Table 5
Profile of the student sample participating in the case study.

age	gender	total (%)	cumulative (%)
young adult (18–44 years old)	female	59.8	59.8
	male	36.7	96.5
middle-age (45–60 years old)	female	2.15	98.7
	male	1.39	100

one (FIS1) is dedicated to analyzing the variables 'processing disorders,' 'language disorders,' and 'mental disorders.' The output is the level of 'learning limitation,' categorized as very small, small, moderate, large, and very large (Fig. 3).

In this process, linguistic labels (low, medium, and high) are assigned to the input variables. Guided by a rule basis (Table 1), the degrees of membership for these labels are then utilized in the inference process to derive an assessment of the student's learning limitation.

Likewise, FIS2 follows a parallel structure, incorporating the input variables 'self-regulation strategies,' 'cognitive strategies,' and 'emotional strategies' (Fig. 4). The result of this integration provides a comprehensive assessment of 'learning potential.

During this inference process, governed by the rule basis (Table 2), each input variable undergoes a parallel membership assignment to linguistic labels, quantifying their individual contributions to the overall learning potential. The resultant 'learning potential' is then categorized into one of the predefined linguistic labels, aiding informed decision-making and the development of tailored educational strategies based on the specifics of the assessed potential.

The outputs from the two preceding subsystems are brought together in the third and final system (FIS3), guided by the rule base shown in Table 3. This integration culminates in the generation of educational approach recommendations that holistically factor in all the previously analyzed variables, shaping a comprehensive outlook on students' learning profiles (Fig. 5).

Upon obtaining the defuzzied FIS3 outputs in crisp real values, this integration resulted in a DSS for intervention actions that aim at improving the educational approach, including recommendations for teaching techniques and assessment methods that are most appropriate for each learning condition (limitation and potential), as can be seen in Table 4.

By combining the insights from these two critical dimensions—learning limitations and potential—the FIS3 provides a comprehensive understanding of students' individual profiles, which allows crafting educational approaches aligned with the multifaceted characteristics exhibited by each student, transcending a one-size-fits-all approach.

4. Application of the proposed framework

In this case study, we delve into the practical implementation of the Fuzzy DSS. The empirical dataset was sourced from an engineering undergraduate population. The data collection was carried out using an online self-reporting form, which encompassed the study strategies, learning disabilities, and academic performance indicators. The survey link was distributed through official university communication channels, fostering a representative sample, providing a total of 822 responses. The process adhered to ethical considerations, with participants providing informed consent (Table 5). Fig. 6 shows how much the learning disabilities hinders the student's performance.

Fig. 7 depicts the density of employing study strategies, which illustrates the extent to which students utilize these strategies in their learning endeavors, providing insights into the patterns of strategy adoption and their potential impact on academic performance.

By using this dataset (Figs. 6 and 7), the ANN model with the highest accuracy during machine learning was attained through a tuning process. This configuration entailed the utilization of 200 neurons distributed across hidden layers. The logistic function was selected as the activation function, to boost the model's ability to model complex relationships within the variables. The L-BFGS-B solver was employed for optimization, effectively navigating the model toward a convergence point that minimized error. Regularization was implemented to prevent overfitting, with an alpha value set to 7.10^{-4}. With a commitment to model convergence and stability, a maximum of 300 iterations was set for the training process. This ensemble of parameters culminated in a MLP that achieved an impressive R^2 value of 0.938, indicating a strong

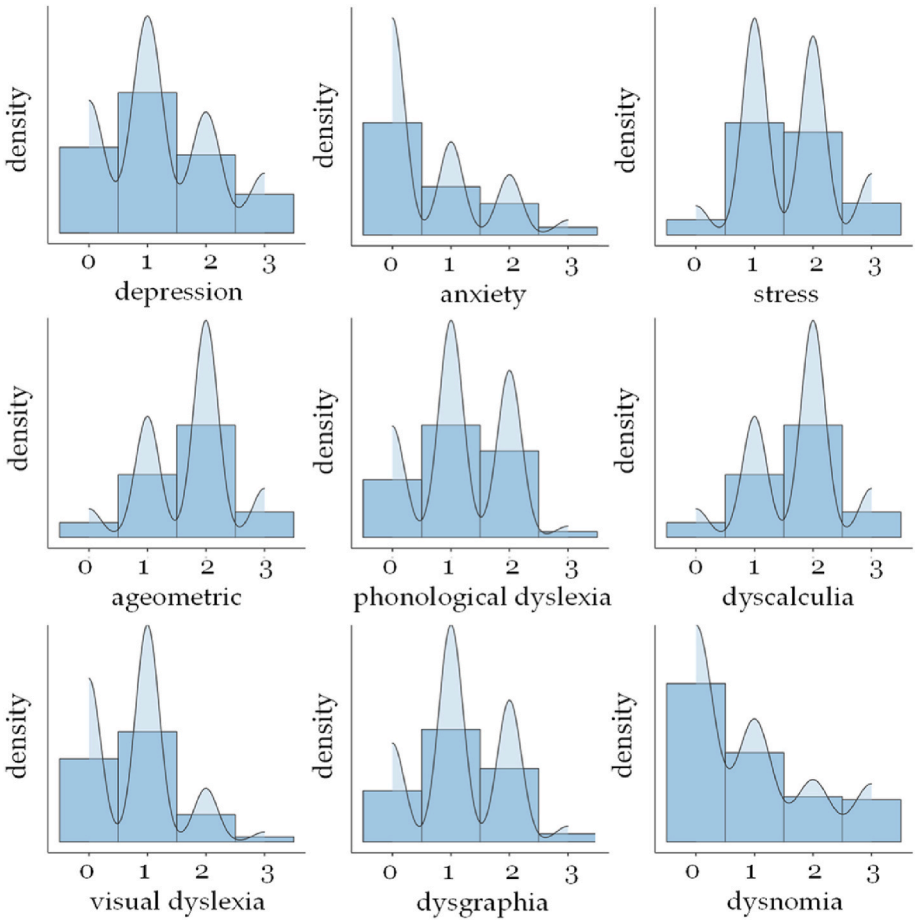


Fig. 6. Occurrence of learning disabilities: 0 – never, 1 – rarely, 2 – often, and 3 – always.

fit between the model's predictions and the observed academic performance. Fig. 8 shows the decrease in the R^2 associated with the removal of each variable, indicating the most influential ones driving academic performance and their relative weight (w). Before entering the FIS, the values declared in the form by the student regarding the frequency of study strategies (Fig. 7) and the occurrence of learning disabilities (Fig. 6) are first pondered by these weights (Fig. 8).

Through this process, the proposed AI-based framework is fine-tuned on a case-by-case basis to the specific characteristics of the student cohort for which the educational approach will be recommended. This calibration is rooted in recognized patterns through the analysis of the historical empirical database, as well as the distinct profile of each student, particularly in terms of the study strategies they consistently employ.

Illustrating with an example based on a respondent's answers from the survey, let's consider the following declared values for the frequency at which learning disabilities negatively impact academic performance: phonological dyslexia - never (0), dyscalculia - often (2), ageometric difficulties - always (3). These values sum up to a total of 5 for the processing disorders. However, when incorporating the weights determined by the ANN, the weighted value for Dyscalculia would be 2 (declared value) multiplied by 0.415 (weight assigned to this variable), and for Ageometric difficulties, the weighted value would be 3 multiplied by 0.424. This would result in a cumulative weighted value of 2.102 for the Processing Disorders. This process is repeated for all input variables within the FIS. Each input value provided by the respondent is adjusted based on the respective weights assigned by the neural network. For this same respondent, the weighted values of the variables led FIS1 to yield a 'moderate' level of learning limitation, while FIS2 indicated a 'large' learning potential. The integrated analysis within FIS3 consequently resulted in the recommendation group labeled as '8.' In essence, for this particular student, the optimal educational approach would involve fostering a collaborative learning environment enriched with opportunities for peer mentoring.

In terms of teaching techniques, the recommendation suggests the utilization of technology-enhanced learning platforms along with group problem-solving activities. These methods aim to foster active engagement and collaborative learning experiences, aligning with the student's recognized learning potential and preferences. Furthermore, the recommended assessment methods for this student align with the identified learning potential and strategies. These assessment strategies combine the use of quizzes, debates, and collaborative presentations. By assessing knowledge application through these diverse methods, educators can better capture the student's comprehensive understanding of the material and their ability to apply it in various contexts.

In sum, the intricate interplay between the three FIS enables the generation of a tailored educational approach for this student. This approach aligns with the student's learning limitations, potential, and study strategies, encompassing collaborative learning environments, technology-enhanced methods, and diversified assessment techniques. This individualized recommendation underscores the study's overarching goal of providing precise and effective educational strategies that cater to the unique learning profiles of each student.

Applying the proposed DSS to a cohort of 35 first-year students in an incoming class yielded distinct sets of recommendations from FIS3: set '5' was recommended to 19 students, set '7' to 7 students, set '4' to 7 students, and set '2' to 2 students. This stratification ensures that students with similar learning profiles are grouped together in class, facilitating the implementation of focused and targeted educational strategies. By categorizing students based on their recommendation sets,

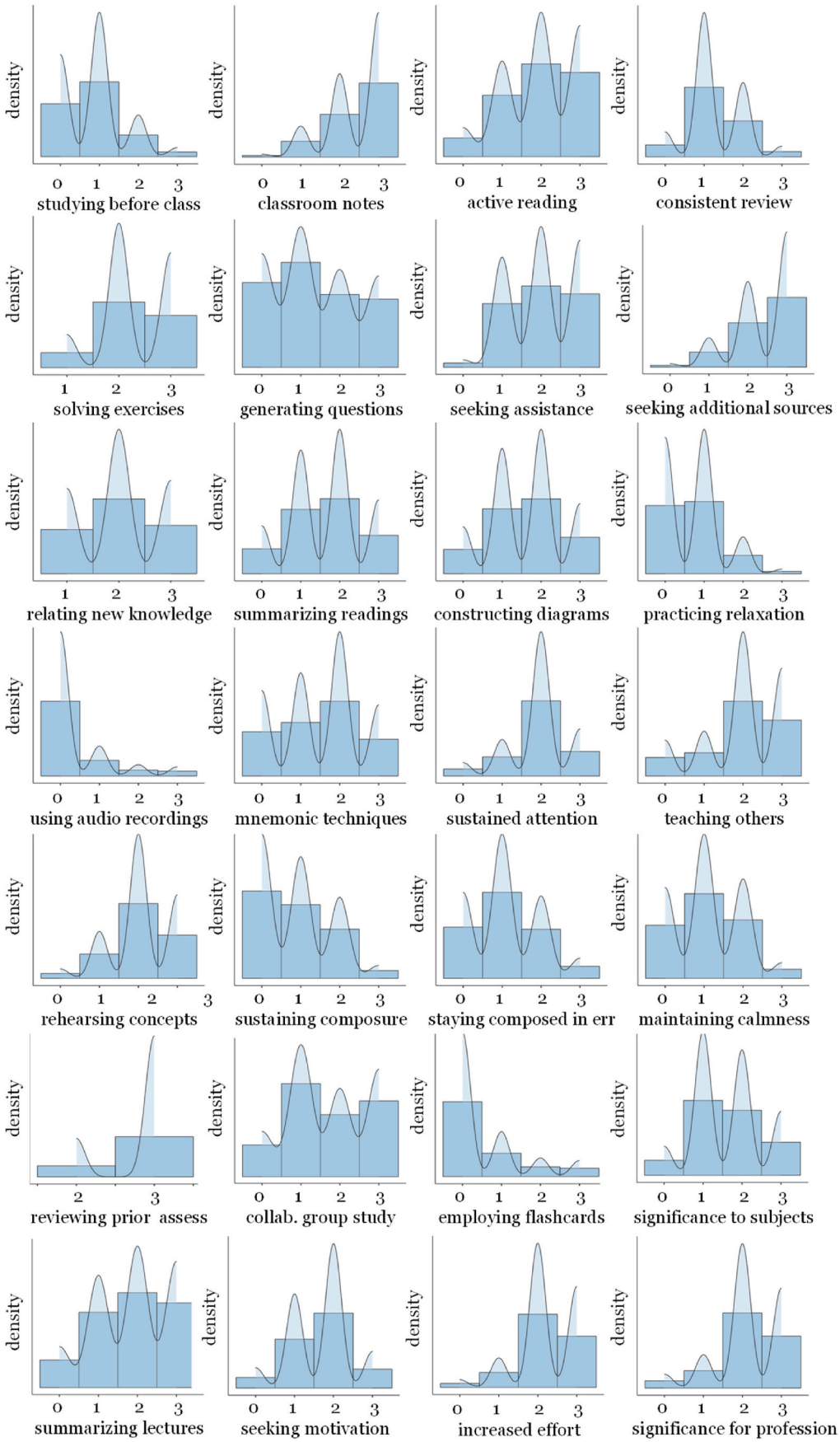


Fig. 7. Use of study strategies by the students: 0 – never, 1 – rarely, 2 – often, and 3 – always.

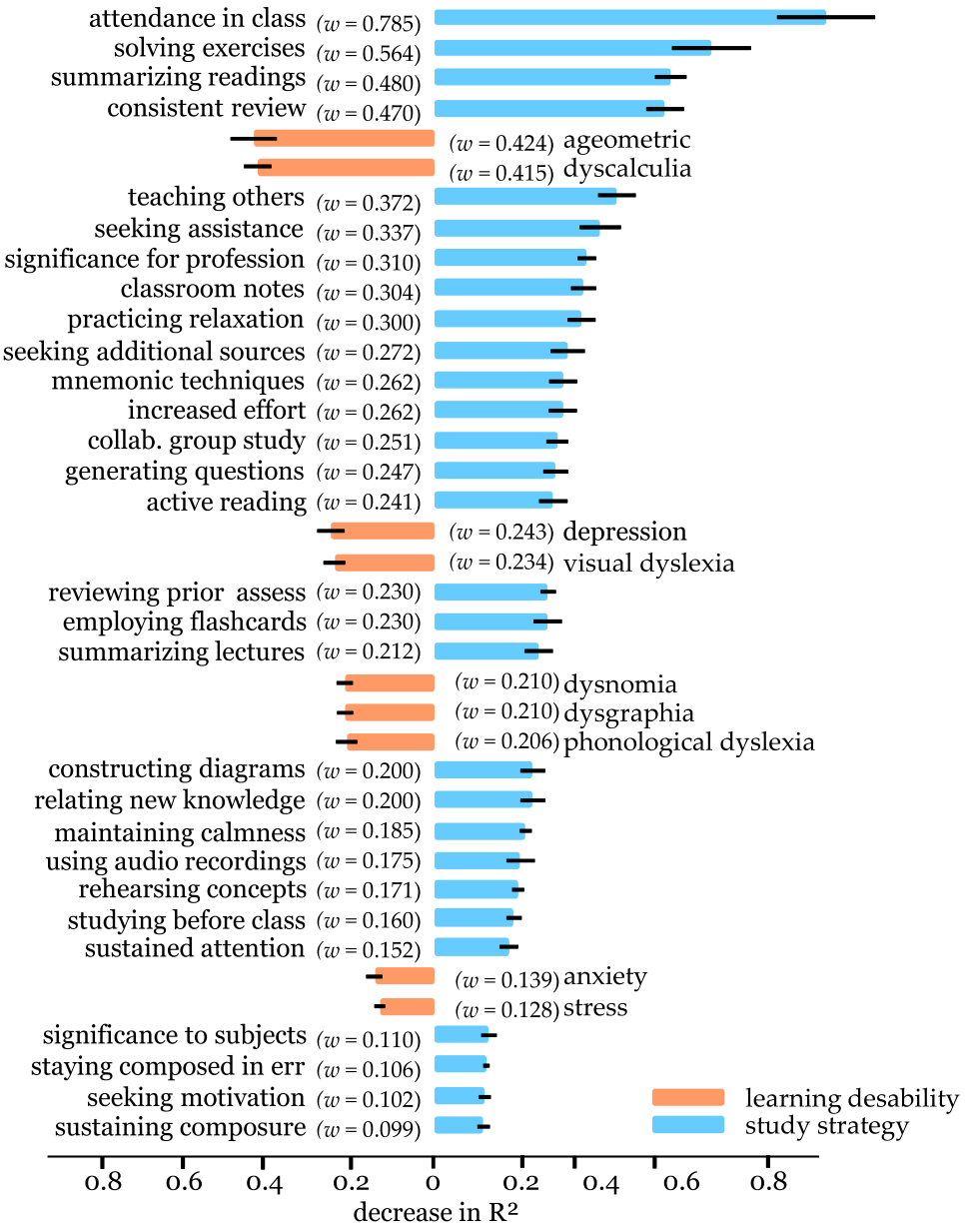


Fig. 8. Coefficient of determination analysis for weighting variable contribution.

educators can streamline their approach, tailoring their teaching techniques, resources, and assessments to the specific needs and preferences of each group. This approach can optimize the learning experience for each student and maximize the potential for academic success within the class.

5. Concluding remarks

Returning to the initial research question, our results allow us to conclude that AI-integrated study strategies can mitigate the impact of learning disabilities on academic performance by providing personalized interventions, adapting educational approaches to students' cognitive profiles, and fostering inclusivity, resulting in improved academic outcomes. The existing landscape of traditional educational methods often falls short in accommodating the diverse needs of students, particularly those with learning disabilities. This research endeavors to bridge this gap by providing a comprehensive understanding of how study strategies and AI-based DSS can work synergistically to optimize educational outcomes. By employing the proposed AI framework, educators can make informed decisions to customize educational approaches that align with the unique cognitive profiles of students. Through personalized interventions guided by recognized patterns, the academic journey becomes more inclusive and effective, benefiting all learners regardless of their cognitive backgrounds or learning challenges.

The practical implications of this research are significant for educators, policymakers, and administrators. The proposed data-driven strategies offer a blueprint for enhancing teaching methodologies, fostering an inclusive learning environment, and promoting academic success for students with learning disabilities. By embracing personalized interventions based on students' learning limitations and potential, educational institutions can foster a more equitable educational landscape, where individual strengths are nurtured, and challenges are addressed effectively.

In a rapidly evolving educational landscape, where diversity and inclusivity are paramount, this study's insights and methodologies pave the way for transformative approaches to teaching and learning. By harnessing the power of AI and leveraging the nuanced relationships

between study strategies, learning disabilities, and academic performance, educators can embark on a journey of continuous improvement, ensuring that each student's learning experience is optimized for success. Ultimately, this research contributes to the advancement of educational practices that prioritize individual growth, inclusivity, and excellence in the pursuit of academic achievement. For future studies, there are some directions that can further expand and improve the proposed framework. Comparing the effectiveness of traditional teaching methods with the AI-driven educational recommendations can be carried out through longitudinal studies, providing insights into how study strategies evolve over time and their impact on academic performance, especially for students with learning disabilities. By delving into this direction, future research endeavors can contribute to a deeper understanding of how study strategies and learning disabilities intersect, and how AI-based tools can be harnessed to create more effective, inclusive, and personalized educational experiences for all students.

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Data availability

Ethical restrictions: due to the nature of this research, participants of this study did not agree for their data to be shared publicly, so supporting data is not available.

Declaration of using generative AI

During the preparation of this work the authors used ChatGPT by OpenAI, July 20 Version, in the writing process to improve language and readability. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

CRediT authorship contribution statement

Adriano Bressane: Conceptualization, Formal analysis, Investigation, Methodology, Project administration, Supervision, Validation, Writing – original draft, Writing – review & editing. **Daniel Zwirn:** Formal analysis, Investigation, Methodology, Writing – original draft, Writing – review & editing. **Alexei Essiptchouk:** Conceptualization, Investigation, Methodology, Writing – original draft, Writing – review & editing. **Antônio Carlos Varela Saraiva:** Formal analysis, Investigation, Validation, Writing – original draft, Writing – review & editing. **Fernando Luiz de Campos Carvalho:** Formal analysis, Methodology, Validation, Writing – original draft, Writing – review & editing. **Jorge Kennety Silva Formiga:** Formal analysis, Validation, Writing – original draft, Writing – review & editing. **Líliam César de Castro Medeiros:** Conceptualization, Methodology, Validation, Visualization, Writing – original draft, Writing – review & editing. **Rogério Galante Negri:** Formal analysis, Validation, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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